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Dynamic Node Selection for Two-Root Polynomials

Modified Secant Method and Chebyshev-Frobenius Eigenvalue Approach

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Abstract

Dynamic Node Selection is an advanced numerical technique developed for solving two-root polynomial systems efficiently and accurately. Classical root-finding methods often experience instability when roots are closely spaced or when only a single initial estimate is available. This report studies a hybrid computational approach that combines the Modified Secant Method with the Chebyshev-Frobenius companion matrix method.

The modified secant method is first applied using a single starting point to estimate the root separation dynamically. Based on this estimated separation, the polynomial interval is refined and transformed into the Chebyshev interval $[-1, 1]$. The transformed polynomial is then expanded using Chebyshev basis functions. A Chebyshev-Frobenius companion matrix is constructed whose eigenvalues provide approximations to all polynomial roots.

The method combines the local convergence properties of iterative techniques with the global spectral accuracy of Chebyshev methods. Dynamic node selection improves numerical stability, minimizes interpolation errors, reduces oscillations, and enables accurate computation of clustered roots. The report also discusses interval transformation, Chebyshev nodes, companion matrix formulation, spectral convergence, algorithmic implementation, and comparison with conventional numerical methods.

Keywords: Dynamic Node Selection, Modified Secant Method, Chebyshev Nodes, Spectral Methods, Companion Matrix, Root Finding.

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1. Introduction

Root finding is one of the fundamental problems in numerical analysis and computational mathematics. Many engineering systems, optimization problems, vibration studies, control systems, fluid mechanics models, and machine learning algorithms eventually reduce to solving polynomial equations.

Traditional numerical methods such as the Newton-Raphson method, Secant method, and Bisection method are effective for simple isolated roots. However, these methods may become unstable or computationally inefficient when roots are clustered closely together. Small root separation creates convergence difficulties and increases numerical sensitivity.

To overcome these limitations, spectral numerical methods based on Chebyshev polynomials have gained significant importance. Chebyshev methods provide excellent numerical stability and minimize interpolation errors. Combining iterative methods with spectral approaches creates powerful hybrid techniques capable of accurately computing all roots of a polynomial.

This report focuses on Dynamic Node Selection for two-root polynomials. The technique combines the Modified Secant Method with the Chebyshev-Frobenius eigenvalue approach. The modified secant method is first used to estimate root separation dynamically using a single initial guess. The interval is then transformed into the Chebyshev domain where spectral methods are applied to compute all roots efficiently.

The research paper studied also investigates the dynamical behavior of Chebyshev's method applied to polynomials with exactly two distinct roots and analyzes the associated Julia and Fatou sets.

2. Objective

The primary objective of this report is to study Dynamic Node Selection for two-root polynomial systems using a hybrid numerical framework.

The detailed objectives are:

1. To understand the limitations of traditional root-finding techniques.
2. To study the Modified Secant Method for estimating roots using a single initial guess.
3. To dynamically estimate root separation for clustered polynomial roots.
4. To transform the polynomial interval into the Chebyshev interval $[-1, 1]$.
5. To study Chebyshev nodes and their numerical advantages.
6. To construct the Chebyshev-Frobenius companion matrix.
7. To compute polynomial roots using eigenvalue analysis.
8. To analyze convergence, stability, and spectral accuracy.
9. To compare Dynamic Node Selection with conventional numerical methods.

3. Two-Root Polynomial Systems

A two-root polynomial contains exactly two distinct roots. A general quadratic polynomial can be expressed as:

$$p(z) = (z - \alpha)(z - \beta)$$

where:

- α and β are distinct roots,
- $\Delta = |\beta - \alpha|$ represents root separation.

Root separation plays an important role in numerical root-finding methods. When roots are very close to each other, traditional iterative methods may converge slowly or even diverge. This phenomenon is called clustered root behavior.

Dynamic node selection addresses this issue by adaptively estimating the root separation and refining the computational interval accordingly.

4. Modified Secant Method

The Modified Secant Method is an iterative numerical technique used to approximate roots without directly computing derivatives.

The iteration formula is:

$$x_{n+1} = x_n - \frac{\delta x_n p(x_n)}{p(x_n + \delta x_n) - p(x_n)}$$

where:

- x_n is the current approximation,
- δ is a small perturbation parameter,
- $p(x)$ is the polynomial function.

The method begins from a single initial point x_0 and performs repeated iterations until convergence is achieved.

Convergence criterion:

$$|p(x_k)| < \epsilon$$

where ϵ is the tolerance error.

4.1 Advantages of Modified Secant Method

- Requires only one initial guess.
- Avoids derivative computation.
- Faster than classical secant method.
- Effective for nonlinear equations.
- Useful for dynamic root estimation.

4.2 Root Separation Estimation

After several iterations, an approximate root $\hat{r}$ is obtained. The root separation estimate is represented as:

$$\hat{\Delta} \approx |\alpha - \beta|$$

This estimated separation is used to refine the computational interval:

$$[a, b] = [\hat{r} - 1.5\hat{\Delta}, \hat{r} + 1.5\hat{\Delta}]$$

This interval refinement significantly improves spectral accuracy.

5. Chebyshev Spectral Method

5.1 Interval Transformation

The refined interval $[a, b]$ is transformed into the standard Chebyshev interval $[-1, 1]$ using affine mapping:

$$t = \frac{2(z - a)}{b - a} - 1$$

Inverse transformation:

$$z(t) = \frac{b - a}{2}(t + 1) + a$$

This transformation improves numerical conditioning and stability.

5.2 Chebyshev Nodes

Chebyshev nodes are defined as:

$$t_k = \cos\left(\frac{(2k - 1)\pi}{2N}\right)$$

where:

- $k = 1, 2, \dots, N$
- N is the number of nodes.

5.3 Importance of Chebyshev Nodes

Chebyshev nodes provide several numerical advantages:

- Minimize interpolation error,
- Reduce oscillatory behavior,
- Avoid Runge phenomenon,
- Improve spectral convergence,
- Enhance numerical stability.

5.4 Chebyshev Polynomial Expansion

The transformed polynomial is represented using Chebyshev basis functions:

$$q(t) = \sum_{j=0}^{N-1} c_j T_j(t)$$

where:

- $T_j(t)$ are Chebyshev polynomials,
- c_j are spectral coefficients.

Chebyshev-Frobenius Companion Matrix

The Chebyshev-Frobenius companion matrix converts the root-finding problem into an eigenvalue problem.

The matrix form is:

$$C = \begin{pmatrix} \frac{c_0}{2c_{N-1}} & 1 & 0 & \cdots & 0 \\ \frac{c_1}{2c_{N-1}} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \frac{c_{N-3}}{2c_{N-1}} & 0 & \cdots & 0 & 1 \\ \frac{c_{N-2}}{2c_{N-1}} & 0 & \cdots & 0 & 0 \end{pmatrix}$$

The eigenvalues of this matrix approximate the polynomial roots.

5.5 Root Recovery

If λ_k are eigenvalues of matrix C , then:

$$\lambda_k \approx \text{roots of } q(t)$$

The roots in the original interval are recovered using:

$$z_k = \frac{b-a}{2}(\lambda_k + 1) + a$$

This transformation converts spectral roots back to physical roots.

6. Dynamic Node Selection Algorithm

The complete algorithm is summarized below:

1. Input polynomial $p(z)$ and initial estimate x_0 .
2. Apply Modified Secant iterations.
3. Obtain approximate root $\hat{r}$.
4. Estimate root separation $\hat{\Delta}$.

5. Refine interval:

$$[a, b] = [\hat{r} - 1.5\hat{\Delta}, \hat{r} + 1.5\hat{\Delta}]$$

6. Transform interval into $[-1, 1]$.

7. Generate Chebyshev nodes.

8. Compute Chebyshev coefficients.

9. Construct companion matrix.

10. Compute eigenvalues.

11. Map eigenvalues back to original roots.

7. Results and Analysis

Dynamic Node Selection provides highly accurate root approximations even for clustered roots. The modified secant pre-processing stage improves interval localization while Chebyshev spectral methods improve global accuracy.

7.1 Major Observations

- Faster convergence compared to conventional methods.
- Improved handling of clustered roots.
- Better numerical stability.
- Reduced interpolation oscillations.
- High spectral accuracy.
- Efficient computation of all roots.

7.2 Comparison with Conventional Methods

gray!15 Method	Advantages	Limitations
Newton-Raphson	Fast convergence	Requires derivative computation
Secant Method	No derivative required	Sensitive to initial estimates
Durand-Kerner	Computes all roots	Less stable for clustered roots
Dynamic Node Selection	High spectral accuracy and stability	Larger matrix computations

7.3 Error Analysis

Chebyshev spectral methods provide exponential convergence:

$$Error = O(2^{-N})$$

where N represents the number of Chebyshev nodes.

This exponential decay makes the method highly accurate for complex polynomial systems.

8. Findings

The major findings obtained from this study are:

1. Dynamic node selection improves root localization accuracy.
2. Modified secant iteration effectively estimates root separation.
3. Adaptive interval refinement improves computational efficiency.
4. Chebyshev nodes reduce interpolation errors significantly.
5. Spectral methods provide high numerical stability.
6. Companion matrix eigenvalues accurately compute all roots.

7. Hybrid iterative-spectral techniques outperform many classical methods for clustered roots.
8. Exponential spectral convergence improves overall accuracy.

9. Conclusion

Dynamic Node Selection is an advanced hybrid numerical strategy developed for solving two-root polynomial systems efficiently and accurately. The method combines the local convergence capability of the Modified Secant Method with the global spectral accuracy of Chebyshev polynomial techniques.

The modified secant stage dynamically estimates root separation using only a single initial estimate. The interval transformation and Chebyshev node selection improve numerical conditioning and minimize interpolation errors. The Chebyshev-Frobenius companion matrix converts the root-finding problem into an eigenvalue problem, allowing efficient computation of all polynomial roots.

The study demonstrates that Dynamic Node Selection provides better stability, improved convergence, and higher computational accuracy compared to many conventional root-finding methods. The method is especially effective for clustered roots and complex polynomial systems.

Future developments may include:

- Extension to higher-degree polynomials,
- Adaptive spectral refinement,
- Multi-dimensional root systems,
- Parallel eigenvalue computation,
- GPU-accelerated spectral algorithms.

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